Grassroots Organisations: Building Digital Banks, Networks, Corporations and Cooperatives

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ABSTRACT

Grassroots platforms are owned, operated and governed by their participants. Each is specified by a grassroots contract, a few clauses in plain English stating who may do what and with whose consent, and is compiled from the contract into a program that its participants run on their smartphones, with no operator in the middle. Fundamental grassroots platforms include the grassroots social graph, grassroots social networks, child-safe social networks, and grassroots currencies, with higher-level platforms providing for democratic digital communities and "sharing-economy" cooperatives.

Grassroots contracts are a digital incarnation of a two-centuries old notion of Pierre-Joseph Proudhon. Proudhon envisioned a social contract that can be freely-adopted by any group of people, where groups bound by the same social contract can union by mutual consent, possibly into a single global union. A methodology to compile grassroots contracts to working smartphone apps, with AI, was developed.

Grassroots platforms can provide social, commercial and political functions for natural persons who are legally-responsible for their acts and trust others to act responsibly. However, there are many other types of agents we wish to accommodate in the digital realm, including children and computational agents (AI agents, cloud-based servers who offer a myriad of digital services). Previous works showed how grassroots contracts use the notion of guardians and wards to incorporate such agents.

Here, we extend this approach to include organisations. We begin with an abstract notion of organisation that has members with roles and bylaws, and then extend and instantiate it to specify banks, network organisations, corporations, and cooperatives. In each case we present a grassroots contract, a smartphone app that implements it, and stochastic simulations of its behaviours.

We thus show that the notion of grassroots contracts, augmented with guardians and wards, can be the basis of a thriving digital realm, owned, operated and governed by its participants.

CCS CONCEPTS

• **Computing methodologies** → **Multi-agent systems**; • **Computer systems organization** → **Peer-to-peer architectures**; • **Human-centered computing** → *Social networking sites*; • **Security and privacy** → *Digital signatures*; • **Applied computing** → *Law*; • **Social and professional topics** → *Computing / technology policy*; • **Theory of computation** → Distributed computing models.

KEYWORDS

Grassroots Platforms, Grassroots Contracts, Social Contracts, Digital Speech Acts, Genuine Digital Identity, Social Networks, Smartphone Apps

1 INTRODUCTION

Grassroots contracts [4, 11] are a digital incarnation of a two-centuries old notion of Pierre-Joseph Proudhon [29, Sixth Study]. Proudhon envisioned a social contract that can be freely adopted by any group of people—one could "make a contract with all, as … with some"—where groups bound by the same social contract can union by mutual consent, possibly into a single global union—"each group of citizens … formed by a like contract … could thereafter, and always by a similar contract, agree with every and all other groups". Grassroots contracts are realised by grassroots platforms [20, 34, 36, 42]: smartphone-based person-to-person digital platforms owned, operated and governed by their participants. A grassroots platform is constituted from its participants' smartphones and the grassroots app each participant runs, and requires no global resource but the network: no hardwired servers, no global authority, and nobody's permission to start. Global platforms, centralised like Facebook or decentralised like Bitcoin and Ethereum [27, 52], can have only a single instance, as two instances would clash over the global resources—domain name, port number, or boot nodes—hardwired into their code, and from the perspective of ordinary people all of them are monolithic entities owned, operated and governed by others [37]. Fundamental grassroots platforms include the grassroots social graph [8, 11, 35], grassroots social networks [35], child-safe social networks [39], and grassroots currencies [19, 33, 41, 45], with higher-level platforms providing for democratic digital communities [13, 17, 47, 49] and "sharing-economy" cooperatives (in preparation).

A methodology to compile grassroots contracts to working smartphone apps, with AI, was developed: an abstraction cascade from the contract to the app (Section 2), its correctness proved from the transactions down to GLP and the app built and tested, each layer coded by AI from the paper that describes it [7, 11, 18, 20, 38, 43, 46].

Grassroots platforms can provide social, commercial and political functions for natural persons who are legally-responsible for their acts and trust others to act responsibly. However, there are many other types of agents we wish to accommodate in the digital realm, including children and computational agents (AI agents, cloud-based servers who offer a myriad of digital services). Previous works showed how grassroots contracts use the notion of guardians and wards to incorporate such agents [39, 40].

In this paper, we extend this approach to include organisations. We begin with an abstract notion of organisation that has members with roles and bylaws, and then extend and instantiate it to specify banks, network organisations, corporations, and cooperatives. In each case we present a grassroots contract, a smartphone app that implements it, and stochastic simulations of its behaviours.

We thus show that the notion of grassroots contracts, augmented with guardians and wards, can be the basis of a thriving digital realm, owned, operated and governed by its participants.

Grassroots platforms are presently a concept with mathematical and computational theory [11, 20, 34, 37, 42], working prototypes [7, 18, 43], and tools for their large-scale simulation and testing [44]. Grassroots theory has been developed for a decade without the resources needed to turn it into competitive software, given the not-for-profit, common-good nature of the project. AI is now piercing through this conundrum: the scientific papers serve as the evolving interface between people and AI, which codes from them [43, 46]. Much work remains for grassroots smartphone apps to be competitive with incumbent global platforms; but perhaps now this could be achieved with some AI-months rather than with incalculably-many person-years.

Section 2 recalls the grassroots contracts this paper builds on: the social graph and social network, guardians and wards, grassroots currencies, and grassroots ride-hailing. Section 3 presents GO, the grassroots organisation, a ward whose officers are its guardians and whose bylaws name who decides each of its acts and how. Section 4 presents banks: the mutual credit society, the community bank, the community currency, the commercial bank and the sovereign central bank. Section 5 presents network organisations, whose providers join by holding a stake in its currency and whose customers by accepting it: ride-hailing, an ad network, lodging, a marketplace, gig work and lending, with arbitration. Section 6 presents corporations: the single-person limited-liability company, the trading company and the shareholding corporation. Section 7 presents cooperatives: the workers', the consumers' and the DAO, with liquid delegation. Section 8 presents the apps compiled from the contracts and their simulation, Section 9 related work and Section 10 concludes.

2 BACKGROUND

A grassroots contract [4, 11] is a few clauses in plain language, each describing an act—who takes it, who must agree to it, and what it changes—or binding the agents without describing one. Befriending is an act of two persons, both of whom must agree; ending a friendship is an act of one. The contract is the whole of the platform's design: there is no server, no operator and no account, only the persons who adopted the contract and their smartphones [34, 36]. Each clause opens with its legal modality [11]: ■ *enforced*: no act of agents can breach it; ◆ *attested* by the signatures of the actors: an agent may breach it, and the contract records the breach; ○ *undertaken* by the actors: its performance lies outside the contract, which records only the undertaking. A contract ends with its *pragmas*, which the app built from it follows [11, 18]. The pragmas of every contract are in the repository https://anonymous.4open.science/r/GLP-02A3, and GKK's, GSG's and GSN's in [11].

From the contract to the app. The clauses are written as speech-act schemas [11], which name the roles of an act, which of them must will it, and what each role's state must hold before and comes to hold after; the schemas compile into volition-guarded multiagent atomic transactions [20], each a change to the states of its participants that occurs only if the persons guarding it are willing; the transactions are realised by communicating volitional agents, whose only acts between two agents are discovery and message delivery, and then by a volition-guarded Grassroots Logic Program (vGLP) [18], which is transformed into the platform's user interface and compiled to Grassroots Logic Programs (GLP) [38, 46], implemented in Dart and Flutter [43] and run as a mini-app of the Grassroots Super-App on a person's smartphone [7] (Figure 1). Correctness is proved from the transactions down to GLP [18, 20]; the app is built and tested. Each layer is coded by AI from the paper that describes it; a gap found in coding is repaired in the paper first, and the code follows [43].

What a person sees. Volition elicitation is the rationale for the user interface design [18]. An act a person must will appears on their phone as a question, with the answers the contract allows [18, 20]: Bob is asked whether to be Alice's friend, and answers Accept or Decline. A person's phone holds their own state and nothing else [34–36], their friends, their groups, the messages they sent and received, and draws what it shows from that state, by widgets the grassroots app declares for the platform's questions and for the person's state [7, 18]. A person's phone talks to the phones of the agents they know or come to know, and to nobody else [7, 8, 35]. The screens of Figures 2 and 3 are screenshots of the GSG and GSN mini-apps, each compiled from its grassroots contract down the abstraction cascade [11, 18, 20, 38, 43, 46]. One instance of the grassroots app [7] on an iPhone simulator runs the mini-app of every agent [7, 43]; the agents communicate over channels within it [38], and a screenshot shows one person's screen. All source code and all artefacts, including the grassroots contracts and the artefacts the abstraction cascade produces from them, are in the public repository https://anonymous.4open.science/r/GLP-02A3.

2.1 Grassroots contracts, platforms and apps

Almost two centuries ago Pierre-Joseph Proudhon envisioned a social contract [29, Fourth Study] that is (1) an agreement of man with man; (2) reciprocal; (3) imposing no obligation upon the parties except that which results from their personal promise; (4) subject to no external authority; (5) freely accepted and signed by all the participants; and (6) of the nature of a contract of exchange. *Grassroots contracts* are social contracts meeting these six conditions and two more: (7) people are free to deal with each other; and (8) there is no external register of people, so people can come to know each other either directly or through people they already know [11].

A *digital social contract* is one realised by software the parties run [4]. We call the mathematical object a *protocol* and the software implementing it a *platform*. A protocol is *grassroots* when its instances are independent of one another and of every global resource but the net, and can always coalesce, possibly into a single global instance [34, 36, 37], and a *grassroots platform* is the software implementing a grassroots protocol [11].

Participating in a grassroots platform requires adopting the grassroots contract that defines it, as participating in a global platform requires adopting an End-User Licence Agreement (EULA). A grassroots contract is mutual and symmetric among all participants; an EULA is between each person and the platform operator. To join a grassroots platform, a person signs a one-sided undertaking to keep its grassroots contract towards all its other participants, and presents it to every participant they interact with, so every participant has signed it [11].

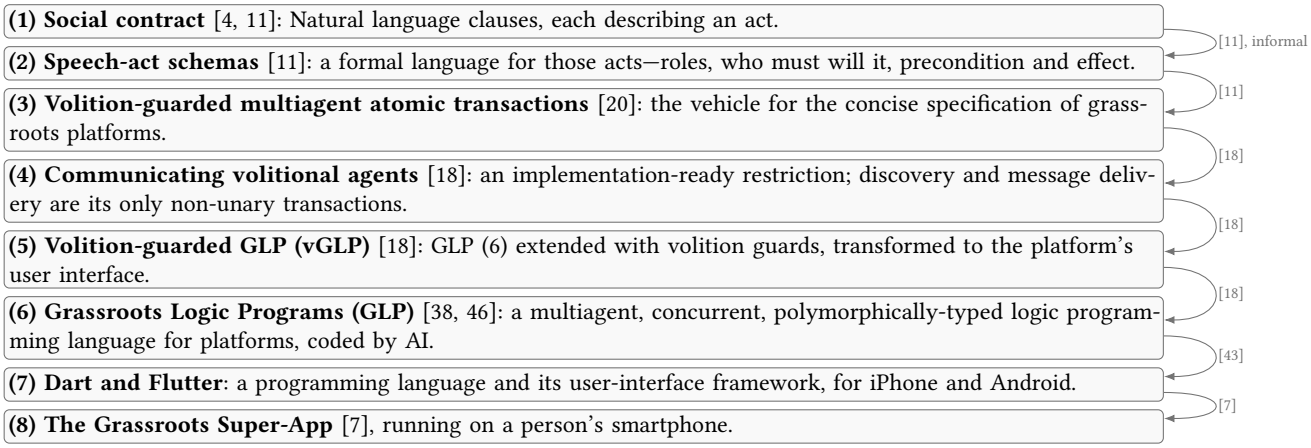


Figure 1: The abstraction cascade, from a grassroots contract to a running app. Each arrow but the first, an informal translation, represents an implementation, with the reference defining it.

2.2 The grassroots social graph and social network

A *natural person* is a human being. A *person* is a legally culpable entity: a natural person of age, also referred to as an *adult*. A person takes the pronoun 'they'. A *juridical person* is an entity other than a human being that the law recognises as holding rights and duties in its own name; a *legal person* is a person or a juridical person. An *agent* is a legal person, a *child*, a natural person not of age, or an *AI agent*, a program that holds a key pair of its own and takes acts of the contract with it, and takes the pronoun 'it'. Every contract extends GKK, the base contract: a person establishes their identity by choosing a key pair, and *knows* another agent when they hold that agent's public key [11].

GKK: Grassroots Keys and Knowledge [11]

Alice, Bob and Carol are persons.

1. ○ *Identity.* Alice may establish her identity once by choosing a unique key pair. She makes herself known by the public key, signs her acts with the private key, and is accountable for every act signed with it.
2. ■ *Acquaintance.* Alice and Bob may come to know each other by an act of both.
3. ■ *Introduction.* If Alice and Bob know each other and Alice knows Carol, then Alice can make Carol known to Bob, with Bob's consent.

A contract that extends another adds clauses to it and cites its clauses whole; GSG extends GKK.

GSG: Grassroots Social Graph [11]

Alice *trusts* Bob when she knows Bob personally and believes him to keep the undertakings he signs under grassroots contracts.

Alice and Bob are persons.

1. ◆ *Friendship.* Alice and Bob, who trust each other and are not friends, may jointly decide to become friends.
2. ■ *Ending a friendship.* Alice may end her friendship with Bob.

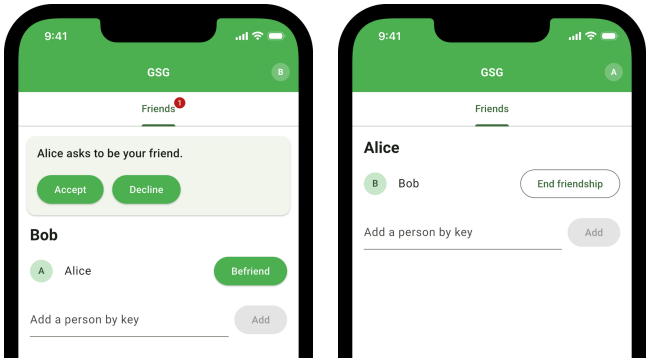


Figure 2: The GSG app: Alice asks to be Bob's friend, and Bob answers Accept or Decline (GSG:1); Alice's friends after Bob accepts.

3. ○ *Loss of Trust.* Alice ends her friendship with Bob when she ceases to trust him.

GSG is the grassroots social graph [11, 35]. The GSG app is a mini-app of the grassroots app, compiled down the cascade from GSG; Figure 2 shows it realising GSG:1.

GSN: Grassroots Social Network [11]

Alice and Bob are persons.

1. ■ *Messaging.* Alice may send a message to her friend Bob.
2. ■ *Group.* Alice may create a group, of which she is the leader and first member; Alice may close the group by leaving it.
3. ■ *Joining and Leaving.* Alice, the leader of a group, may add her friend Bob to it, with Bob's consent; either Alice or Bob may then end Bob's membership in the group, which also ends when Alice and Bob cease to be friends.
4. ■ *Group messaging.* A member of a group may send a message to the group, and its leader delivers it to every current member.

The GSN app is a mini-app of the grassroots app, compiled down the cascade from GSN; Figure 3 shows it realising GSN:2, GSN:3 and GSN:4.

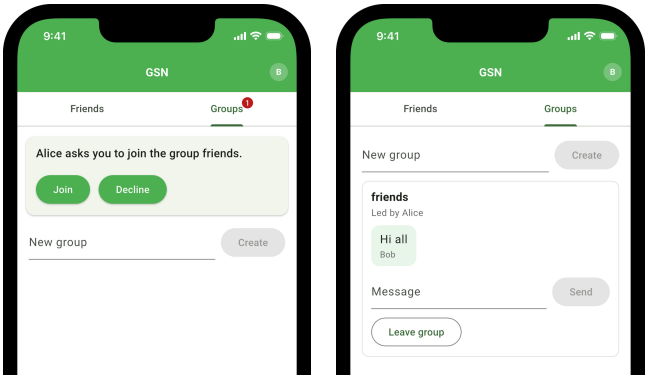


Figure 3: The GSN app, on Bob's phone: Alice asks Bob to join her group *friends*, and Bob answers Join or Decline (GSN:3); the group *friends*, led by Alice (GSN:2), with Bob's message to the group as Alice delivered it to him (GSN:4).

REC: Recommendations

REC extends GKK [11].
Alice, Bob and Carol are persons.

1. ◆ *Recommending.* Alice may recommend Bob, whom they know, for something, a skill or a service, once for each thing; Bob is not asked and keeps the recommendation.
2. ■ *Showing.* Bob may show a recommendation they hold to Carol, whom they know, once; Carol then sees who recommended Bob and for what, but does not thereby come to know the recommender.
3. ■ *Withdrawing.* Alice may withdraw a recommendation; it is gone from Bob and from everyone Bob showed it to, and Alice may not make it again.

2.3 Guardians and wards

GnW, Guardians 'n' Wards, is the contract of guardians and wards, and a contract with wards is certified under the conditions stated with it [11].

2.3.1 Children. The child-safe social contract, CSSN [39], extends GSN (Section 2.2), and through it GSG and GKK, and GnW (Section 2.3), cites their clauses whole, and has five clauses of its own, for parents and children. It covers the standard functions of social networking—messages between friends, and groups—and child safety.

A child's identity is a *ward*, operated by the child, and a guardian of it is a *parent* of the child.

CSSN: Child-Safe Social Network

Alice and Bob are adults, Carol and Dave children.

1. ◆ *Parent.* Only a legal parent of a child, Alice, may create the ward Carol, operated by the child.
2. ■ *A child's group.* Carol may create a group with the consent of a guardian, Alice; only a child, Dave, may join it.
3. ○ *Coming of age.* The creator of Carol may declare Carol of age, which ends every guardianship of Carol, Carol's friendships remaining, and Carol is then a person; the creator shall do so when Carol comes of age, and not before.
4. ○ *A child with no guardian.* Alice, aware that Dave is a child though not a ward, shall neither befriend Dave nor let their child befriend Dave, and shall report it to Dave's parents if Alice knows who they are.
5. ■ *A child's messages.* Carol may send a message to a friend, and to a group of which they are a member.

Identity, parenthood and majority. A person establishes their identity by choosing their own key pair (GKK:1), and a parent establishes their child's when they create it, becoming the child's first friend and its creator, who may make a friend of theirs its parent (GnW:1, GnW:2 and CSSN:1), so a child is a child from their first act. Whether an adult is a child's parent, and whether a child has come of age, are facts no act of a grassroots contract can determine, there being no global registry and no global clock: the contract records who declared them, and the courts decide whether the declarations are true. Guardianship ends only by an act of a parent, and the creator's declaration of age ends every guardianship and leaves the child's friendships in place, so a child cannot leave child safety at will.

Community-Specific Social Contracts. A child-safe social contract need not be universal. The contract is written at the level on which parents and educators discuss how children may meet, play and learn together, and each community may adopt its own, to its values, its culture and its approach to education, and revise it as its children grow; each such contract is a separate mini-app of the grassroots app, derived from its own clauses as this one is.

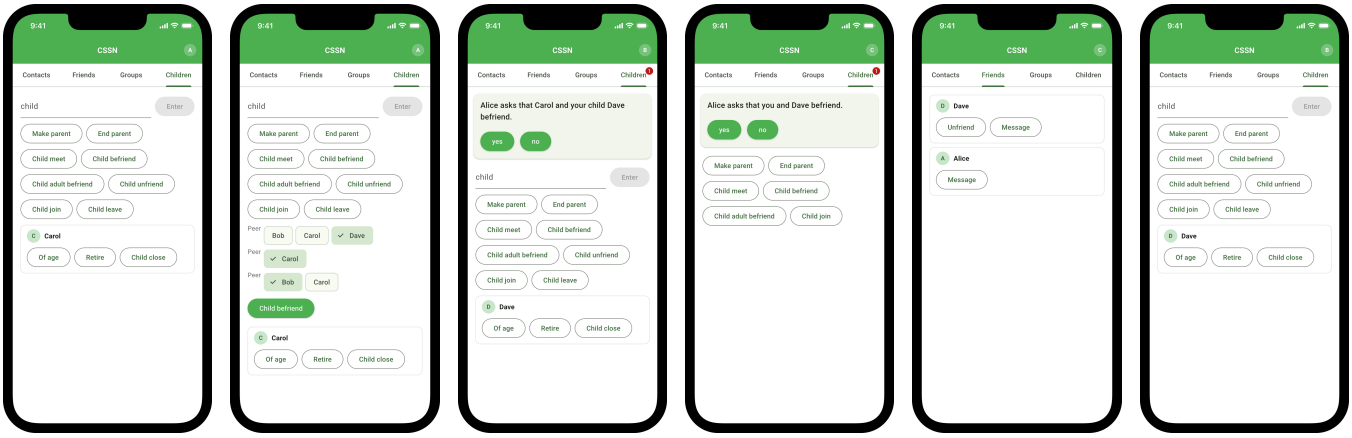
The child-safe platform runs as a mini-app of the grassroots super-app [7], which knows nothing of the relation between a parent and their child, it being clauses and guards in the mini-app's own program. The parent's approval is asked on the parent's own phone, and the child's act does not occur until the parent answers it. Figure 4 shows the mini-app in a scenario of four people: Alice and Bob, friends, request that their children Carol and Dave come to know each other and then befriend, the other three approving; Carol ends the friendship, and Bob declares Dave of age.

2.3.2 AI agents. An AI agent is made by a person, its *developer*, and is operated by its guardians, among them the person who created it, its *principal*. Accountability for an AI agent is divided between the two as it is for a car between its manufacturer and its driver: the manufacturer is accountable for manufacturing defects, and the driver for reckless driving. So the developer signs the code of the AI agent, and its principal attests, in a speech act signed over the AI agent's public key, that the AI agent runs that code, and every act of the AI agent carries the attestation.

GAI: Accountable AI Agents

GAI extends GnW and amends GFP [11].
Alice is a person and Alf an AI agent.

1. ○ *AI developer and guardian accountability.* The *developer* of an AI agent, Alf, signs Alf's code, and is accountable for Alf's defects. Every guardian of Alf is accountable for operating it recklessly. The creator of Alf is its *principal*.



(1) (2) (3) (4) (5) (6)

Figure 4: The child-safe mini-app in the grassroots app, on the iPhone simulator, in the scenario of Section 2.3.1, one person's screen each: (1) Alice's Children tab, after she and Bob come to know each other and befriend: the field to enter a child, the buttons of the acts that name two or more, and her child Carol, with the acts that name only Carol; (2) her request that Carol and Dave befriend, after the one that they come to know each other, Carol, Dave and Bob picked into its places; (3) Bob asked to approve it, with yes and no; (4) Carol asked to approve it; (5) Carol's Friends, Dave, with Unfriend and Message, and Alice, her creator and first friend, with Message, whom she may not unfriend; (6) Bob's Children tab after Carol ends the friendship, with Of age beside Dave, Bob being Dave's creator. Every word on a screen is the contract's, but yes and no.

2. ■ *AI acts.* Every act of Alf names its developer and its principal, and carries its principal's attestation, signed over Alf's public key, that Alf runs code signed by its developer.

In GAI the wards are AI agents and their guardians the persons who operate them. The principal's will is given once, when they create the AI agent; after that the AI agent acts on its own, and a guardian takes part in each friendship it enters (GnW:3 and GnW:4): an AI agent comes to know an agent by an act of the two, and befriends one only with a guardian. The principal signs the attestation once too, when they create the AI agent, and the AI agent presents it to each agent it comes to know, who records it (GAI:2). A principal may make other persons guardians of an AI agent (GnW:2). A guardian signs the undertaking to keep the contract for their ward [11].

Figure 5 shows the contract on a person's phone, in the GSNP app: a person creates an AI agent, attesting that it runs code signed by its developer, approves each friendship of the AI agent, and may retire it; every agent an AI agent knows sees its principal and its developer.

2.3.3 Servers. A *store* is a ward of its owner, as a rendezvous server is (GNA, in preparation): an agent with a key of its own, run on a machine, holding a list of programs; what a person does with a store is the contract GAS, which extends GPD [7] and GnW, the contract of guardians and wards [40]; the text in the box is the repository's copy, `programs/social/graph/GAS.txt`. The *Grassroots App Store*, at https://peoplesweb.org, is the store of the GLP repository, in its directory `servers/peoplesweb`; the site, the server and the files that deploy them are in the repository, and anyone can set up a store (GAS:1) and run one from them.

GAS: Grassroots App Store

GAS extends GPD and GnW [40].
A *store* is a ward operated by a machine at an address every person can reach, holding a list of programs.
Alice, Bob and Carol are persons, Dave a ward.

1. ■ *Store.* Carol may create a store, Dave, as her ward.
2. ○ *Listing.* Carol may list in Dave a program developed by Bob, after adding her signature to its signatures, as long as she trusts Bob as a developer; and may delist it.
3. ■ *Store Trust.* Alice may trust Dave, Carol being known to her, and may withdraw her trust.
4. ■ *Installation from a Store.* Alice may install a program listed in a store she trusts.

Trust in a store is trust in its owner's choice of programs: the owner lists a program while they trust its developer, signing it (GAS:2), so the store's owner, and not each person, judges the developers; a person who trusts no store installs from the developers they trust themselves (GPD:4).

The store's agent. The store's agent is compiled from GAS and runs on the server over the networking layer with no person and no screen, under the ward's key, which the site shows with its owner's. A person comes to know the store by that key (GKK:2), by a cold call over the Internet [7], which the store answers for anyone; the store is nobody's friend, and its channel is unattested, as a rendezvous server's is [7]. A program may travel over an unattested channel, its integrity resting on its signatures and not on the channel. The Grassroots App Store's owner is the developer of the programs of the repository, publishing each (GPD:1) and listing it (GAS:2), so a person who trusts the store (GAS:3) installs any of them (GAS:4).

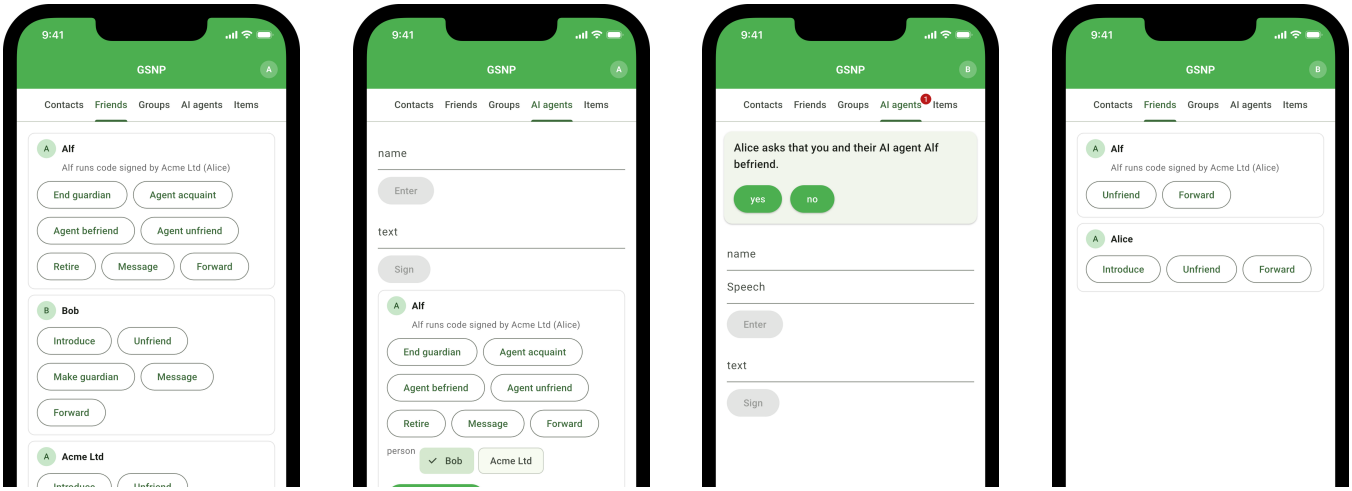


Figure 5: Accountable AI agents, in the GSNP app: Alice's AI agent Alf, its code signed by Acme Ltd and attested by her, and its Retire button (GnW:1, GAI:2 and GnW:5); Alice proposes that Alf befriend Bob (GnW:3); Bob is asked to befriend Alf, an AI agent of Alice (GAI:2); Bob's friends, Alf among them with its principal and its developer.

2.4 Grassroots currencies

GC, GB and DGC, the contracts of grassroots coins, bonds and denominated currencies [33, 41, 45], with persons or organisations as their parties.

GC: Grassroots Coins [33]

GC extends GKK.
Alice and Bob are persons or organisations.

1. ■ *Minting.* Alice may decide to issue Alice-coins.
2. ■ *Swap.* Alice and Bob, who know each other, may jointly decide to swap coins they hold.
3. ■ *Payment.* Alice may decide to pay Bob-coins she holds to Bob, whom she knows.
4. ■ *Redemption.* Alice may decide to redeem a Bob-coin she holds against any coin held by Bob, whom she knows.
5. ○ *Pricing.* Alice prices her offerings in Alice-coins and accepts them in payment at those prices.

GB: Grassroots Bonds [41]

GB extends GC.
Alice, Bob and Carol are persons or organisations.

1. ■ *Minting.* Alice may decide to issue Alice-bonds of a maturity later than her own date.
2. ○ *Date.* Alice keeps her date at the current date.
3. ■ *Swap.* Alice and Bob, who know each other, may jointly decide to swap coins and bonds they hold.
4. ■ *Redemption.* Alice may decide to redeem a Bob-coin, or a matured Bob-bond, she holds against any coin or bond held by Bob, whom she knows.
5. ■ *Escrow.* Alice may deposit coins and bonds with Carol for Bob, on a condition the three agree; Carol and Bob may jointly decide to release them to Bob when the condition is met, and Carol and Alice may jointly decide to return them to Alice when it fails; each act is between two parties who know each other.

DGC: Denominated Grassroots Currencies [45]

DGC extends GB.
Dollar names one denomination, and an Alice Dollar-denominated coin or bond, Alice-coin and Alice-bond for short, is one issued by Alice.
Alice, Bob and Carol are persons or organisations.

1. ■ *Minting.* Alice may decide to issue their coins, of any denomination, and their bonds, of any denomination and of a maturity later than their own date.
2. ○ *Date.* Alice keeps their date at the current date.
3. ■ *Swap.* Alice and Bob may jointly decide to swap Dollar-denominated coins and bonds they hold.
4. ■ *Acceptance.* Alice may decide to accept Carol-coins in payment up to a limit of their choosing, if they hold a Carol-coin or a Carol-bond; each Carol-coin paid to Alice counts against the limit, and Alice may decide again.
5. ■ *Payment.* Alice may decide to pay Bob Dollar-denominated coins they hold that Bob accepts.
6. ■ *Redemption.* Alice may decide to redeem a Bob-coin they hold against any Dollar-denominated coin or bond Bob holds, and to take a Bob-coin for a matured Bob-bond they hold.
7. ○ *Pricing.* Alice prices their offerings in Alice-coins and accepts them in payment at those prices.
8. ○ *Denomination.* An Alice-coin and an Alice-bond are each Alice's debt of one Dollar, the coin due now and the bond at its date; Alice redeems an Alice-coin on presentation at face value, by a Dollar-denominated coin or bond the bearer accepts or by one Dollar.
9. ■ *Escrow.* Alice may deposit Dollar-denominated coins and bonds with Carol for Bob; Carol and Bob may jointly decide to release them to Bob, and Carol and Alice to return them to Alice, as the agreed condition is met or fails.

2.5 Grassroots ride-hailing

GDO, GDO-LB, GRA and GRA-SS, the contracts of the grassroots directed overlay, its location-based restriction, the grassroots ride auction and its second-score restriction (GRH, Grassroots Ride-Hailing, in preparation).

GDO: Grassroots Directed Overlay (GRH, Grassroots Ride-Ha

GDO extends GSG.
Alice, Bob and Carol are persons.

1. ■ *Pointing to friend.* Alice, who does not point to her friend Bob, may point to Bob.
2. ■ *Transitive pointing.* Alice, who knows Carol and points to Bob who points to Carol, may point to Carol.
3. ■ *Ceasing to point.* Alice, who points to Bob, may cease pointing to Bob.

GDO-LB: Location-Based GDO (GRH, Grassroots Ride-Hailing

GDO-LB restricts GDO.
Alice, Bob and Carol are persons. A person has a location. $d > 0$ is an integer and $e > 0$.

1. ■ *Pointing to friend.* GDO:1, only if either Alice points to less than d persons and Bob is closest to Alice among all of Alice's unpointed friends, or Bob is closer to Alice by more than e than some person pointed to by Alice.
2. ■ *Transitive pointing.* GDO:2, only if Carol is closer to Alice by more than e than some person pointed to by Alice.
3. ■ *Ceasing to point.* GDO:3, only if Alice points to more than d persons, with Bob furthest from Alice among them.

GRA: Grassroots Ride Auction (GRH, Grassroots Ride-Hailing in preparation)

GRA extends GDO-LB.
Alice is a rider, Bob a driver and Carol a person. A person can be either a rider or a driver but not both.

1. ■ *Call.* Alice, who has no open call and no ride, may call for bids on a ride from the persons Alice points to and the persons they point to, stating the pickup point, the destination, the bidding period and the terms.
2. ■ *Bid.* Bob, who received the call, may bid a fare and a pickup delay before the bidding period ends; Alice may not open a bid before then.
3. ■ *Award.* When the bidding period ends, Alice awards the ride to the bidder the terms select among the bidders who have no ride, at the fare the terms set, and Alice's fare and the deposit the terms set for Bob are then locked; if the terms select no bidder, the call is closed.
4. ■ *Pickup.* The pickup is recorded when Alice and Bob are both at the pickup point.
5. ■ *Arriving.* When Alice and Bob are both at the destination, the fare and the deposit are released to Alice and to Bob as the terms say.
6. ■ *Cancelling.* If no pickup is recorded within the maximum pickup delay, Alice may cancel, and the fare and the deposit are released to Alice and to Bob as the terms say.
7. ■ *Withdrawing.* Alice may withdraw her open call before the bidding period ends.

GRA-SS: Grassroots Ride Auction, Second-Score Restriction (GRH, G

GRA-SS restricts GRA.
Alice is a rider, Bob a driver and Carol a person. The terms of a call state a maximum pickup delay $T \geq 0$, a willingness to pay $W(\tau) = A - \lambda\tau$ with $A, \lambda \geq 0$, and a deposit $D \geq 0$. A pickup delay is measured from the award, at time t_0. A bid (p, τ) is the fare $p \geq 0$ the bidder asks and the pickup delay $\tau \geq 0$ the bidder promises. Its score is $W(\tau) - p$, and it is admissible iff $\tau \leq T$ and its score is at least 0.

1. ■ *Transitive pointing.* GDO-LB:2, where Carol is a driver.
2. ■ *Award.* GRA:3, where the terms select the bidder whose admissible bid has the highest score, ties broken as the terms say, or no bidder if no bid is admissible. They set the fare $P = W(\tau) - s_2$, where τ is that bid's pickup delay and s_2 is the highest score among the other admissible bids, or 0 if there is none, and they set the deposit D.
3. ■ *Arriving.* GRA:5, where the refund $\min\{P, \lambda\delta\}$ is released to Alice, and the rest of the fare and the deposit to Bob, with $\delta = \max\{0, t_{\text{pickup}} - t_0 - \tau\}$.
4. ■ *Cancelling.* GRA:6, where the fare and the deposit are released to Alice if Alice was at the pickup point from t_0 to $t_0 + T$, and otherwise the fare to Alice and the deposit to Bob.

3 GRASSROOTS ORGANISATIONS

GO: Grassroots Organisation

GO instantiates GnW [11] and extends it.
An *organisation* is a ward with *participants*, which include its guardians, referred to as its *officers*, each with a *role*. The state of the ward holds its *bylaws* and its *register*, recording the participants and the officers in each role. The bylaws define the roles of the organisation, each with its powers and the acts by which a participant is appointed to it and removed from it, and name for each act of the organisation the officers and/or persons who decide it and the rule by which they decide it jointly, including admitting and removing a participant, and appointing an officer when there is none. Alice and Bob are persons, Orga a ward.

1. ■ *Founding.* Alice, or an organisation with Alice, may create an organisation, Orga, as a ward, choosing its bylaws, with Alice its first officer.
2. ■ *Acts of the organisation.* An officer, or a person the bylaws name, may propose an act of Orga, and the officers and/or persons the bylaws name for it may vote on it; an officer the bylaws name for the act may take it for Orga once their votes decide it according to the bylaws.
3. ■ *Participation.* Bob becomes a participant of Orga by an act of Bob and Orga; his participation ends by an act of Bob or of Orga, and with it every role he holds.

4. ■ *Officers.* A participant may be appointed to a role by an act of Orga and the participant, or removed from it by an act of Orga or of the participant, and is a guardian of Orga while they hold a role.
5. ■ *Amendment.* The bylaws may be amended by an act of Orga.
6. ○ *Accountability.* Orga is accountable at law for its undertakings where it is a juridical person, and every officer who decided an act of Orga is accountable for it with Orga.

Orga is retired by its officers jointly (GnW:5).

4 GRASSROOTS BANKS

GBank: Grassroots Bank

GBank extends GO and DGC.
A *bank* is an organisation whose business is issuing coins and bonds against the coins and bonds it holds.

4.1 Mutual credit society

MCS: Mutual Credit Society

MCS instantiates GBank and amends it.
A *mutual credit society* is a bank with one role, *member*, held by every participant, whose acts are decided by all its members jointly. A member can be *in the red* by their coins in circulation less the other persons' coins they hold.
Alice and Bob are persons, Orga a mutual credit society.

1. ■ *Founding.* Alice may found Orga and become its first member.
2. ■ *Membership.* Bob becomes a member of Orga by an act of Bob and of the members of Orga; his membership ends by an act of Bob or of the other members.
3. ■ *Redemption.* Bob, a member, may redeem an Alice-coin he holds, Alice a member, against an Orga-coin.
4. ■ *Credit limit.* A member can be in the red by at most 1000 coins.
5. ■ *Leaving.* Upon ending membership, the Orga-coins Bob holds and the Bob-coins Orga holds are redeemed against each other.

A society of N members has liquidity of at least $1000N$ Orga-coins, which members increase by mutual credit, swapping their coins with one another (GC:2). Orga can go bankrupt: a member who leaves in the red leaves Orga holding their coins and no coin to redeem them with, so the Orga-coins outstanding exceed the coins Orga holds, and Orga is bankrupt while its members are not.

4.2 Community bank

CB: Community Bank

CB instantiates GBank and restricts it.
A *community bank* is a bank with two roles, *member*, held by every participant, and *director*, held by members the members elect; the directors jointly decide the acts of the bank, and the members jointly elect and remove directors. A member's *credit line* is a number of coins the directors set.
Alice and Bob are persons, Orga a community bank.

1. ■ *Founding.* Alice may found Orga and become its first member and director.
2. ■ *Membership.* Bob becomes a member of Orga by an act of Bob and the directors; his membership ends by an act of Bob or of the directors.
3. ■ *Directors.* The members may elect a member a director, and remove a director.
4. ■ *Credit line.* The directors may set Bob's credit line.
5. ■ *Credit.* Bob may swap Bob-coins he holds for Orga-coins, as long as the Bob-coins Orga holds do not exceed his credit line.
6. ○ *Community currency.* A member shall accept Orga-coins in payment for their offerings at the prices they post.
7. ■ *Leaving.* Upon ending membership, the Orga-coins Bob holds and the Bob-coins Orga holds are redeemed against each other.

The community bank is GC's [33]: its credit lines with its members, the members' coins as backing, and risk management by redemption are described there.

4.3 Community currency

CC: Community Currency

CC instantiates GBank and restricts it.
A *community currency* is the Dollar-denominated coins of an organisation, Orga, with one role, *treasurer*; the treasurers jointly decide the acts of Orga.
Alice and Bob are persons, Orga a community currency.

1. ■ *Founding.* Alice may found Orga for the Dollar and become its first treasurer.
2. ■ *Treasurers.* Bob becomes a treasurer of Orga by an act of Bob and the treasurers; he ceases to be one by an act of Bob or of the other treasurers.
3. ■ *Drawing.* Alice, a treasurer, may swap Alice-coins for Orga-coins.
4. ○ *Fiat redemption.* Bob may redeem an Orga-coin he holds against one Dollar from any treasurer.

The backing is the treasurers' coins Orga holds, against which a holder redeems an Orga-coin (DGC:6); a holder is no participant, and accepts Orga-coins by DGC:4. The community currency is that of GCC, Grassroots Community Currencies, in preparation, with the coins issued by Orga under its key in place of the treasurers' coins declared as the community's, so they stay valid when the treasurers change, and backed as there:

GBCC: Grassroots Backed Community Currency

GBCC extends CC and restricts it.
f is a fraction, $0 \le f < 1$. Orga's *account* is a bank account in Dollars, held in Orga's name by its treasurers. A treasurer, Alice, is *backed* when the Alice-coins Orga holds are at most the Dollars Alice paid into the account and the coins Alice

holds of the other treasurers, leaving out the fraction f of them of whom she holds the most.
Alice and Bob are persons, Orga a community currency.

1. ■ *Treasurers.* CC:2, only if every treasurer stays backed.
2. ■ *Drawing.* CC:3, only while Alice stays backed.
3. ◆ *Account.* Alice, a treasurer, may pay Dollars into the account, and the treasurers jointly record them as paid in by Alice.
4. ○ *Fiat redemption.* CC:4, from the account while it holds Dollars, and from a treasurer when it holds none.

GCC's result carries over: an Orga-coin is redeemed for a Dollar while at most a fraction f of the treasurers are insolvent or removed; above f, Orga can go bankrupt. The account is outside the contract: a payment into it and a redemption from it are acts the treasurers undertake, and the contract records them.

4.4 Commercial bank

CMB: Commercial Bank

CMB instantiates GBank and restricts it.
A *commercial bank* is an organisation that may issue Dollar-denominated coins and bonds, with two roles, *director* and *loan officer*; the directors jointly decide the acts of the bank; a loan officer decides a loan within their *lending limit*, a number of Dollars the directors set, and the directors decide a loan above it. A participant who holds no role is a *customer*.
Alice, Bob and Carol are persons, Orga a commercial bank.

1. ■ *Founding.* Alice may found Orga and become its first director.
2. ■ *Customers.* Bob becomes a customer of Orga and ceases to be one by an act of Bob and the directors.
3. ■ *Officers.* The directors may appoint a participant a director or a loan officer, and remove one.
4. ■ *Deposit.* Bob, a customer, may swap Dollar-denominated coins he holds for Orga-coins.
5. ■ *Loan.* Bob, a customer, and Carol, a loan officer, for a loan within Carol's lending limit, or the directors, for any loan, may jointly decide to swap Orga-coins for Bob-bonds.
6. ■ *Lending limit.* The directors may set the lending limit of Carol, a loan officer.

A deposit is Bob's claim on Orga in Orga-coins, and a withdrawal is his redemption of them against the Dollar-denominated coins Orga holds (DGC:6), its *reserve*; a loan gives Bob Orga-coins against his bonds, the *interest* being the face of the bonds above the coins given, and when the bonds mature Orga redeems them (DGC:6). Orga is solvent while it holds at least as many Dollars as it owes, and a run on it is met exactly then [33].

4.5 Sovereign central bank

SCB: Sovereign Central Bank

SCB instantiates GBank and restricts it.
A *central bank* is an organisation that is the sovereign of a denomination, Dollar: its Dollar-denominated coins and bonds are the *sovereign coins* and *bonds* of the Dollar. It has two roles, *governor* and *committee member*; the governors jointly decide the acts of the bank, and the committee members, the *monetary policy committee*, set by a majority the *policy rate* for each term, the return over that term at which the bank lends its coins against bonds. A participant who holds no role is a *counterparty*.
Gov is an organisation, Alice and Bob persons, Orga a central bank.

1. ■ *Founding.* Gov, with Alice, may found Orga as the sovereign of the Dollar, with Alice its first governor.
2. ■ *Counterparties.* Bob becomes a counterparty of Orga and ceases to be one by an act of Bob and the governors.
3. ■ *Officers.* The bylaws name the officers of Gov as the persons who decide the appointment and the removal of a governor or a committee member.
4. ■ *Policy rate.* The committee members may set the policy rate for a term.
5. ■ *Lending.* Bob, a counterparty, and Orga may jointly decide to swap Orga-coins for Bob-bonds of a term, as many bonds as the coins and the interest on them at the policy rate for that term.
6. ■ *Open market.* Bob, a counterparty, and Orga may jointly decide to swap coins and bonds they hold.
7. ○ *Authority.* Gov founds Orga and appoints and removes its officers as the law of the Dollar provides.

Clause 5 is the policy rate of SGC [45]; a loan at a penalty rate to a distressed counterparty, the lender of last resort, is clause 6 at the terms Orga sets; and the reserve of a commercial bank (CMB) is the sovereign coins it holds. Who the counterparties are, banks or any person, is the bank's mandate [45].

5 GRASSROOTS NETWORK ORGANISATIONS

GNO: Grassroots Network Organisation

GNO extends GBCC and restricts it.
A *network organisation* is a backed community currency, Orga, with four roles: *treasurer*, as GBCC; *provider*, held by Bob while his *stake*, the Orga-coins he holds, is at least S, a number of Orga-coins the treasurers set; *customer*, held by whoever accepts Orga-coins; and *arbiter*. A treasurer is a provider. A *dispute* is over Orga-coins locked between a provider and a customer by an act of the organisation.
Alice, Bob and Carol are persons, Orga a network organisation.

1. ■ *Stake.* The treasurers may set S.
2. ■ *Treasurers.* GBCC:1, where Bob is a provider.
3. ■ *Arbiters.* The treasurers may appoint Carol an arbiter, and remove Carol.

4. ■ *Arbitration.* Alice or Bob may raise a dispute to an arbiter, Carol, who then decides the release of the locked coins to Alice and to Bob, less the fee the terms set, which is released to her.

No act admits a provider or a customer; the stake is the provider's membership and the acceptance the customer's. A treasurer draws Orga-coins against their backing (GBCC:2) and sells them to a provider, who holds them as his stake and is paid in them; a holder redeems an Orga-coin for a Dollar from a treasurer (CC:4) or against a treasurer's coin Orga holds (DGC:6). A provider who is not a treasurer risks his stake and backs nothing; the treasurers back the currency and set S, so the network organisation grows by providers at no risk to its backing. An arbiter decides a dispute presence cannot (GNO:4), and a provider who is not at the place or does not provide loses his deposit to the customer. What a provider provides, and what a customer pays, is the extension's: rides, ads, lodging, goods, work and loans below.

5.1 Ride-hailing

ORH: Ride-Hailing Organisation

ORH extends GNO and amends GRA-SS (GRH, Grassroots Ride-Hailing, in preparation).
A *driver* is a provider and a *rider* a customer.
Alice and Bob are persons, Orga a ride-hailing organisation.

1. ■ *Driving.* GRA-SS, where Bob bids and drives as a driver while his stake is at least S, the fare and the deposit in Orga-coins, and the deposit locked on top of his stake.
2. ■ *Riding.* GRA-SS, where Alice calls as a rider while she accepts Orga-coins and holds the fare.

5.2 Grassroots ad network

GAN: Grassroots Ad Network

GAN extends GNO and GDO (GRH, Grassroots Ride-Hailing, in preparation).
An *advertiser* is a provider and a *member* a customer. A member's *profile* is in their state and nowhere else. An *ad* names the profile it is for, the Orga-coins its advertiser pays for a view and for a click, the most one member may earn from it, and its *budget*, the most it pays in all.
Alice and Bob are persons, Orga an ad network.

1. ■ *Posting.* Bob, an advertiser, may post an ad to the members he points to, its budget locked on top of his stake, and a member may pass it to the members they point to.
2. ■ *Showing.* Alice, a member, is shown an ad she received if it is for her profile.
3. ◆ *Viewing and clicking.* Alice may attest that she viewed, or clicked, an ad shown to her, and the ad's budget then pays her its coins for the view or the click, up to the most one member may earn from it.

Nobody learns a profile: the ad goes to every member it reaches, and Alice's phone shows it to her only if it is for her profile (GAN:2). A view and a click are attested, so Bob's loss to a false attestation is bounded by what one member may earn from the ad; the ad's budget, locked at Orga, pays the members, and the treasurers earn by selling the coins advertisers buy as their stake.

5.3 Lodging

GNL: Network Lodging

GNL extends GNO, GDO (GRH, Grassroots Ride-Hailing, in preparation) and REC.
A *host* is a provider and a *traveller* a customer. A *listing* names a place, the nights offered and the price a night in Orga-coins; its *terms* say what is released to whom on cancellation.
Alice and Bob are persons, Orga a lodging organisation.

1. ■ *Listing.* Bob, a host, may post a listing to the travellers he points to, and a traveller may pass it to the travellers they point to; Bob may withdraw it.
2. ■ *Booking.* Alice, a traveller, and Bob, who know each other, may jointly book nights of his listing; Alice's payment for them and the deposit the terms set for Bob, on top of his stake, are then locked.
3. ■ *Check-in and checkout.* Alice may record her check-in when she is at the place on the first night, and her checkout when she leaves it after the last; the payment and the deposit are then released to Bob.
4. ■ *Cancelling.* Alice or Bob may cancel a booking before the first night, and the payment and the deposit are released to Alice and to Bob as the terms say; if no check-in is recorded on the first night while Alice is at the place, she may cancel and both are released to her.
5. ■ *Dispute.* Alice, checked in, or Bob may raise a dispute over the payment and the deposit before checkout (GNO:4).
6. ◆ *Review.* REC:1, where Alice recommends Bob for the stay after checkout.

The host's deposit and the traveller's payment are locked and released as GRA's fare and deposit are; the traveller records her presence at the place (GNL:3), so a host who is not there loses his deposit (GNL:4), and a place not as listed goes to an arbiter (GNL:5). A host is recommended for a stay only by a traveller who stayed (GNL:6).

5.4 Marketplace

GNM: Network Marketplace

GNM extends GNO, GDO (GRH, Grassroots Ride-Hailing, in preparation) and REC.
A *seller* is a provider and a *buyer* a customer. A *listing* names a good, its price in Orga-coins and how it is delivered; its *terms* say what is released to whom on cancellation.
Alice and Bob are persons, Orga a marketplace.

1. ■ *Listing.* Bob, a seller, may post a listing to the buyers he points to, and a buyer may pass it to the buyers they point to; Bob may withdraw it.
2. ■ *Order.* Alice, a buyer, and Bob, who know each other, may jointly order the good; Alice's payment and the deposit the terms set for Bob, on top of his stake, are then locked.

3. ◆ *Delivery.* Alice may attest that the good was delivered to her, and the payment and the deposit are then released to Bob.
4. ■ *Cancelling.* Alice or Bob may cancel the order before delivery, and the payment and the deposit are released to Alice and to Bob as the terms say.
5. ■ *Dispute.* Alice or Bob may raise a dispute over the payment and the deposit before they are released (GNO:4).
6. ◆ *Review.* REC:1, where Alice recommends Bob for the good after delivery.

Delivery is attested by the buyer, since no presence records it; a buyer who received the good and does not attest it keeps the seller's coins locked until the seller raises a dispute (GNM:5), and the arbiter decides.

5.5 Gig work

GNG: Network Gig Work

GNG extends GNO, GDO (GRH, Grassroots Ride-Hailing, in preparation) and REC.
A *worker* is a provider and a *client* a customer. A *job* names the work, its price in Orga-coins and its deadline; its *terms* say what is released to whom on cancellation.
Alice and Bob are persons, Orga a gig-work organisation.

1. ■ *Job.* Alice, a client, may post a job to the workers she points to, and a worker may pass it to the workers they point to; Alice may withdraw it.
2. ■ *Engagement.* Alice and Bob, a worker, who know each other, may jointly engage Bob for the job; Alice's payment and the deposit the terms set for Bob, on top of his stake, are then locked.
3. ◆ *Delivery.* Bob may attest that he delivered the work to Alice.
4. ■ *Acceptance.* Alice may accept delivered work, and the payment and the deposit are then released to Bob; she may reject it, raising a dispute (GNO:4).
5. ■ *Cancelling.* Alice or Bob may cancel the engagement before delivery, and the payment and the deposit are released to Alice and to Bob as the terms say; if no delivery is attested by the deadline, Alice may cancel and both are released to her.
6. ◆ *Review.* REC:1, where Alice recommends Bob for the work after acceptance.

The deadline takes the place of presence: a worker who does not deliver by it loses his deposit (GNG:5); work delivered and rejected goes to an arbiter (GNG:4).

5.6 Lending

LIB: Grassroots Library

LIB extends GB [41].
Alice and Bob are persons.

1. ◆ *Shelving.* Alice may put a thing of their own on their shelf, under a name they have not used before.
2. ■ *Lending.* Alice and Bob, who know each other, may jointly decide that Alice lends Bob a thing from Alice's shelf until a day later than Alice's date; it leaves the shelf, and Bob holds it but cannot lend it on.
3. ■ *Returning.* Bob may return a borrowed thing at any time; Alice is not asked, and it goes back on Alice's shelf.
4. ○ *On time.* Bob returns a borrowed thing by the agreed day.

GNLB: Network Lending

GNLB extends GNO, GDO (GRH, Grassroots Ride-Hailing, in preparation), LIB and REC.
An *owner* is a provider and a *borrower* a customer. A *listing* names a thing on the owner's shelf, the fee and the deposit for lending it, in Orga-coins.
Alice and Bob are persons, Orga a lending organisation.

1. ■ *Listing.* Alice, an owner, may post a listing to the borrowers she points to, and a borrower may pass it to the borrowers they point to; Alice may withdraw it.
2. ■ *Lending.* LIB:2, where Bob is a borrower who received the listing, and Bob's fee and deposit are then locked.
3. ■ *Returning.* LIB:3, upon which the fee is released to Alice and the deposit to Bob.
4. ■ *Late.* If the thing is not returned by the agreed day, Alice may claim the fee and the deposit, which are then released to her.
5. ■ *Dispute.* Alice or Bob may raise a dispute over the fee and the deposit before they are released (GNO:4).
6. ◆ *Recommending.* REC:1, where Bob recommends Alice for the loan, or Alice Bob, after its return.

LIB:4, an undertaking, is enforced in GNLB by the deposit: a borrower who keeps the thing beyond the agreed day loses it to the owner (GNLB:4).

6 GRASSROOTS CORPORATIONS

GCorp: Grassroots Corporation

GCorp extends GO and DGC.
A *corporation* is an organisation that holds, pays and issues Dollar-denominated coins and bonds.

6.1 Single-person limited-liability company

SPLLC: Single-Person Limited-Liability Company

SPLLC instantiates GCorp and restricts it.
A *single-person limited-liability company* is a corporation whose bylaws name one role, the *sole officer*, who decides every act of the company, and admit no participant but the sole officer.
Alice is a person, Orga a single-person limited-liability company.

1. ■ *Founding.* Alice may found Orga, holding its sole office.

2. ■ *Acts of the company.* Orga may take an act by the decision of its sole officer.

Clause 1 fixes the bylaws of GO:1 and clause 2 strengthens GO:2; GO:3 is deleted, and the rest of GCorp stands, so the sole officer may amend the bylaws (GO:5) and make Orga a company of many. Alice founds Orga and decides every act of it; Orga may issue Orga-coins, and Alice may back them by the coins she pays to Orga. An Orga-coin is redeemed against the coins Orga holds (GC:4) and against nothing Alice holds; so Orga may be unable to redeem its coins while Alice is solvent, and Alice's liability is what she paid in: Orga can go bankrupt without bankrupting Alice.

6.2 Trading company

TC: Trading Company

TC instantiates GCorp and restricts it.
A *trading company* is a corporation with three roles, *buyer*, *keeper* and *seller*; its officers jointly appoint and remove officers.
A *good* is a thing Orga buys and sells; Orga *holds* a good from its receipt by a keeper until its release.
Alice, Bob, Carol, Dave and Eve are persons, Orga a trading company.

1. ■ *Founding.* Alice may found Orga and become its first buyer, keeper and seller.
2. ■ *Officers.* The officers may appoint a participant a buyer, a keeper or a seller, and remove one.
3. ■ *Purchase.* Alice, a buyer, and Dave, who know each other, may jointly decide that Orga pays Dave coins for a good Dave delivers to Orga.
4. ◆ *Receipt.* Bob, a keeper, may record the receipt of a good Orga bought, and Orga then holds it.
5. ■ *Sale.* Carol, a seller, and Eve, who know each other, may jointly decide that Eve pays Orga coins for a good Orga holds, which is then sold to Eve and Orga holds it no more.
6. ○ *Keeping.* Bob, a keeper, keeps every good Orga holds, and releases a good sold to Eve to Eve.

A good is sold only while Orga holds it (TC:5), and Orga holds it from a keeper's receipt until its sale (TC:4); the keeper's receipt is attested, so a false receipt is recorded against the keeper. Each officer is accountable for the acts they decided (GO:6): the buyer for the purchase, the keeper for the receipt and the keeping, the seller for the sale. The payments are DGC's (DGC:5), Orga paying and being paid as a person does.

6.3 Shareholding corporation

SC: Shareholding Corporation

SC instantiates GCorp and restricts it.
Share is a denomination, and a *share* of Orga is an Orga-coin denominated in Share. A *shareholding corporation* is a corporation with two roles, *shareholder*, held by every person who holds a share of Orga, and *director*; the shareholders decide by a majority of the shares they hold, and the directors jointly decide the other acts of Orga.
Alice and Bob are persons, Orga a shareholding corporation.

1. ■ *Founding.* Alice may found Orga, which issues a number of shares to her, and become its first director.
2. ■ *Shares.* Orga may issue shares by a decision of the shareholders; Bob is a shareholder while he holds one.
3. ■ *Directors.* The shareholders may elect a participant a director, and remove a director.
4. ■ *Dividend.* The directors may decide a dividend, a number of Dollars a share, a shareholder may then claim it from Orga, once, in Dollar-denominated coins for the shares they hold.
5. ■ *Redemption.* DGC:6, for a coin of a denomination other than Share.

A share is paid as a coin is (DGC:5), to a person who accepts Orga's shares (DGC:4), and is redeemed against nothing (SC:5); a shareholder's return is the dividend. The shareholders' vote is weighted by their shares, a rule of the bylaws (GO:2).

7 GRASSROOTS COOPERATIVES

GCoop: Grassroots Cooperative

GCoop extends GO and DGC.
A *cooperative* is an organisation with two roles, *member*, held by every participant, and *board member*; the members decide by a majority of the members, one member one vote, and the board members jointly decide the acts the members delegate to the board.
Alice and Bob are persons, Orga a cooperative.

1. ■ *Founding.* Alice may found Orga and become its first member.
2. ■ *Membership.* Bob becomes a member of Orga by an act of Bob and the members; his membership ends by an act of Bob or of the other members.
3. ■ *Board.* The members may elect a member to the board, and remove one from it.

7.1 Workers' cooperative

WC: Workers' Cooperative

WC instantiates GCoop and restricts it.
A *workers' cooperative* is a cooperative whose members work for Orga.
Alice and Bob are persons, Orga a workers' cooperative.

1. ■ *Pay.* Orga may pay Bob, a member, Dollar-denominated coins for his work, as the members decide.
2. ■ *Distribution.* The members may decide to distribute Dollar-denominated coins Orga holds, which Orga then pays to the members equally.

7.2 Consumers' cooperative

CnC: Consumers' Cooperative

CnC extends GCoop and TC.
A *consumers' cooperative* is a cooperative that is a trading company whose customers are its members; its *fee* is a number of Dollars the members set, and the *cost* of a good is the coins Orga paid for it.
Alice, Bob and Eve are persons, Orga a consumers' cooperative.

1. ■ *Membership.* GCoop:2, where Bob pays Orga the fee on becoming a member.
2. ■ *Officers.* TC:2, where the members appoint and remove the buyers, keepers and sellers.
3. ■ *Sale.* TC:5, where Eve is a member and pays the cost of the good.

The fees fund the purchases, and a member buys at cost; the producers' cooperative is GCC's farmers', and the drivers' cooperative GRH's, each a community currency whose treasurers are its producers.

7.3 DAO

GDAO: Grassroots DAO

GDAO instantiates GCoop and restricts it.
A *DAO* is a cooperative with one role besides member, *steward*; the steward decides the admission of a member, and the members decide every other act of the DAO by a majority of the members.
Alice and Bob are persons, Orga a DAO.

1. ■ *Founding.* Alice may found Orga and become its steward and first member.
2. ■ *Joining.* Bob becomes a member of Orga by an act of Bob and the steward.

GLDAO: Grassroots Liquid DAO

GLDAO extends GDAO.
A member's *delegate* is another member whose vote counts for theirs.
Alice, Bob and Carol are persons, Orga a DAO.

1. ■ *Delegating.* Bob, a member, may make Carol, a member he knows, his delegate, one at a time, and may revoke it.
2. ■ *Counting.* A member who has not voted on a proposal counts as voting as their delegate did, down a chain of delegates; a member whose chain ends with no vote, or in a cycle, is not counted.

A DAO's treasury is the coins Orga holds, which a member may pay to it (DGC:5), and a grant is a payment by Orga to the proposer of a passed proposal, an act of Orga (GO:2).

8 APPS AND SIMULATIONS

9 RELATED WORK

Smart contracts [48] and decentralised autonomous organisations [2, 9, 51] run on a global blockchain, so every organisation shares one instance and pays its fees; Aragon [6] and MakerDAO [22] are organisations of that kind, governed by token holders [3]. A grassroots organisation runs on its officers' phones, is accountable through its officers (GO:6), and two organisations share nothing but the contract. Platform cooperatives [1, 28, 31, 32] are cooperatives that operate a platform for their members, with the platform's server and its operator; a network organisation has neither, its providers joining by a stake in its currency (Section 5). Gig work on global platforms [26, 30, 50, 53] and its regulation [5, 10] concern the platform's power over its workers; in GNG the workers are the providers and the clients the customers of an organisation whose treasurers back its currency and set its stake. Cooperatives [12, 14, 15] are owned by their members, one member one vote, which GCoop states as its bylaws. Community currencies [16, 21, 23] are kept by an operator; CC is issued by an organisation under its contract, and backed by its treasurers' coins. Money creation by commercial banks and the central bank [24, 25] is CMB's deposit and loan and SCB's lending and open market, as grassroots currencies [33, 41, 45] render them.

10 CONCLUSION

A grassroots organisation is a ward whose guardians are its officers and whose bylaws say who decides each of its acts and how. From GO, by extension, instantiation and restriction, we specified banks, network organisations, corporations and cooperatives, each a grassroots contract of a few clauses, and from the contracts the apps are compiled. The organisations of the digital economy, from a one-person company to a central bank and from a marketplace to a gig-work network, can thus be owned, operated and governed by their participants, with no operator in the middle.

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